



The effects of preoperative aromatherapy massage on anxiety and sleep quality of colorectal surgery patients: A randomized controlled study

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ABSTRACT

Aims: To examine the effects of aromatherapy massage on anxiety and sleep quality in patients undergoing colorectal surgery in the preoperative period.

Background: In recent years, there has been an increase in the number of studies conducted on aromatherapy massage. It is stated that studies conducted on aromatherapy massage for anxiety and sleep quality reveal contradictory results and that more research is required on the issue.

Design: A randomized controlled trial.

Methods: Eighty patients undergoing colorectal surgery were randomly assigned to experimental and control group. To the experimental group (n = 40), aromatherapy massage was applied in accordance with the “Back Massage Guide” using 5% lavender oil (*Lavandula Hybrida*) for ten minutes before surgery and the morning of surgery. The control group received standard nursing care in compliance with the hospital procedure. Data were obtained by the State Anxiety Inventory (SAI) and Richard-Campbell Sleep Questionnaire (RCSQ). Results were analyzed using the t-test, Chi-square test or Fisher’s exact test.

Results: There was no baseline difference between the groups. A statistically significant difference was found between the experimental and control group in terms of the SAI and RCSQ mean scores recorded on the morning of surgery. It was determined that the SAI and RCSQ mean score of the experimental group after aromatherapy massage on the morning of surgery decreased when compared to that of the evening before surgery.

Conclusions: It was found that aromatherapy massage with lavender oil increased the sleep quality and reduced the level of anxiety in patients with colorectal surgery in the preoperative period.

1. Introduction

Recent research has demonstrated that the incidence of colorectal cancer is in an increasing trend and it is likely that the number of colorectal cancer survivors is going to increase in the near future if the advances in detection treatment and surgical technique are taken into consideration.¹ Colorectal cancer is the second most common cancer in women and the third most common cancer in men.²

The incidence of colorectal cancer is rapidly increasing in developed countries. In the United States, 135,430 new colon and rectum cancer cases were estimated to be diagnosed in 2017, and 50,260 people were expected to die of this disease.³ In Europe, 345,000 colorectal cancers were diagnosed in 2012.² In Turkey, colorectal cancers are the third most common cancer type among males (8.3%) and males (9.1%).⁴

In colorectal cancer, the primary treatment is surgical intervention. However, the postoperative recovery is not only affected by surgical procedures, but also by the individual’s health condition and anxiety.⁵ The pre-operative anxiety is one of the common problems faced by

patients and is seen in about 11–80% of patients who are about to undergo surgery.⁶ In a study conducted with patients who underwent colorectal surgery, it was reported that patients with colon and rectum resections experienced anxiety.⁷ At the same time, patients who have concerns and fear about their lives and future in the preoperative period may also experience sleep problems due to anxiety and stress.^{5,8,9}

Reduction of anxiety and enhancing quality of sleep increase patient satisfaction and enable the individual to undergo a successful surgical operation.^{10–12} Today, integrative approaches such as aromatherapy are used in addition to medication to enhance quality of sleep and reduce anxiety.^{13–16} Furthermore, nurses are also responsible for nursing practices involving the use of essential oils supported with evidence-based practices.¹⁷ It has been reported that aromatherapy has many positive effects on psychological factors such as depression, anxiety and stress and physiological factors such as sleep, fatigue, blood pressure, nausea, vomiting, pruritus and pain.^{13,14,16,18–23}

Nurses have certain responsibilities to ensure that patients who undergo colorectal surgery receive a successful operation, provide

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patient satisfaction, and maximize patients' well-being. In this context, reducing the anxiety level of patients and increasing their sleep quality in the preoperative period is an undeniably important part of the holistic nursing approach. In the literature review, it was seen that the studies conducted on aromatherapy massage within the framework of evidence-based approach revealed contradictory results and that there was a need for further research on this subject. Furthermore, a study on the effect of lavender oil massage on anxiety and sleep quality in the preoperative period was found neither in Turkey nor in the world. It is thought that this study will shed light to the literature about the effects of aromatherapy massage using lavender oil on anxiety and sleep quality in the preoperative period.

1.1. The aim of the study

The aim of this study is to the effects of lavender oil aromatherapy massage on the anxiety and sleep quality of patients who will undergo colorectal surgery.

1.2. Hypotheses

H1. In patients undergoing colorectal surgery, aromatherapy massage reduces the level of anxiety in the preoperative period.

H2. In patients undergoing colorectal surgery, aromatherapy massage enhances sleep quality in the preoperative period.

2. Methods

2.1. Design

This study was designed as a randomized controlled trial with pre-test and post-test model. The sample of the research consisted of patients who would undergo colorectal surgery at the general surgery unit of a university hospital in Turkey between January 25th and May 31st, 2016. Persons with the following characteristics were not included in the sample of the research; younger than 18 years old, susceptible to lavender oil, cognitively impaired, presence of mental illness (delirium, delusions), use of antidepressants, anti-histamines, diuretics, hypnotics, benzodiazepines and narcotic derivatives that affect the quality of sleep, presence of any respiratory disorder such as asthma, use of other complementary and integrative practices during treatment, inability to or undesirable conditions for receiving massage and unwillingness to participate in the research. The sample size was determined by performing a power analysis.

In the G-Power statistical program, power analysis was performed using the STAI mean scores before and after the aromatherapy massage group of Wu et al.¹⁴ The sample size for each group was calculated to be 25 in the power analysis based on type 1 error 0.05 and type 2 error 0.20 (80%) power. Eighty patients were included in the study; 40 for the experimental group and 40 for the control group. For the post power analysis, the G-Power 3.0.10 statistical power analysis software was used. Also, the State Anxiety Inventory (SAI) and Richard-Campbell Sleep Questionnaire (RCSQ) were used to calculate the mean scores at the night before the surgery and the morning of the surgery to reveal a total power of 99%.

2.2. Randomization

The individuals to undergo colorectal surgery who were in compliance with the sampling inclusion criteria were determined via electronic patient registration system. Patients on the surgery list of the following day were determined and a selection system was conducted where the first patient on the list was included in the experimental group while the next was included in the control group respectively (Chart 1).

2.3. Interventions

In order to determine the functionality of the prepared questionnaire form and the scales used, a pre-test was performed with a total of 6 patients; 3 from the control and 3 from the experimental group between January 25 and February 5, 2016 in the General Surgery Clinic of the hospital where the study was conducted. After the pre-test, a rearrangement of the forms was not required and the patients who had taken the pre-test were excluded from the sample.

2.3.1. Experimental group

After the groups were determined by the randomization method, each patient in the experimental group was informed about the aim of the study in the patient room between 16.00 and 17.00 h, and after receiving their written approval, the patients were asked to fill in the Individual Identification Form, Richard-Campbell Sleep Questionnaire (RCSQ) and State Anxiety Inventory (SAI). The forms were filled by the patients in approximately ten minutes.

Massage was applied twice by the certified researcher in accordance with the "Back Massage Guide" using 5% lavender oil (Lavandula Hybrida) for ten minutes between 19:00–21:00 h before surgery and 10 min between 06.30–08.00 h on the morning of surgery. After the morning session was over, patients were asked to fill in the RCSQ and SAI again.

The reason for choosing the Lavandula Hybrida essential oil is its chemical structure composed of esters which are sedative and calming.^{24,25,26} Essential oils used in aromatherapy are used after being diluted with fixed oils as their application can irritate the skin.¹⁷ For this reason, sweet almond oil was used to dilute lavender oil. In the research, the experimental group was given a 5% mixture obtained by adding 5 ml of lavender oil to 95 ml almond oil.²⁷

2.3.2. Control group

Each patient in the control group was informed about the aim of the study in the patient room between 16.00 and 17.00 h, and after receiving their written approval, the patients were asked to fill in the Individual Identification Form, RCSQ and SAI. Patients in the control group received standard nursing care in compliance with the hospital procedure applied in the preoperative period. At 06.30–08.00 h on the morning when the patients were scheduled for surgery, they were asked to fill in the RCSQ and SAI again.

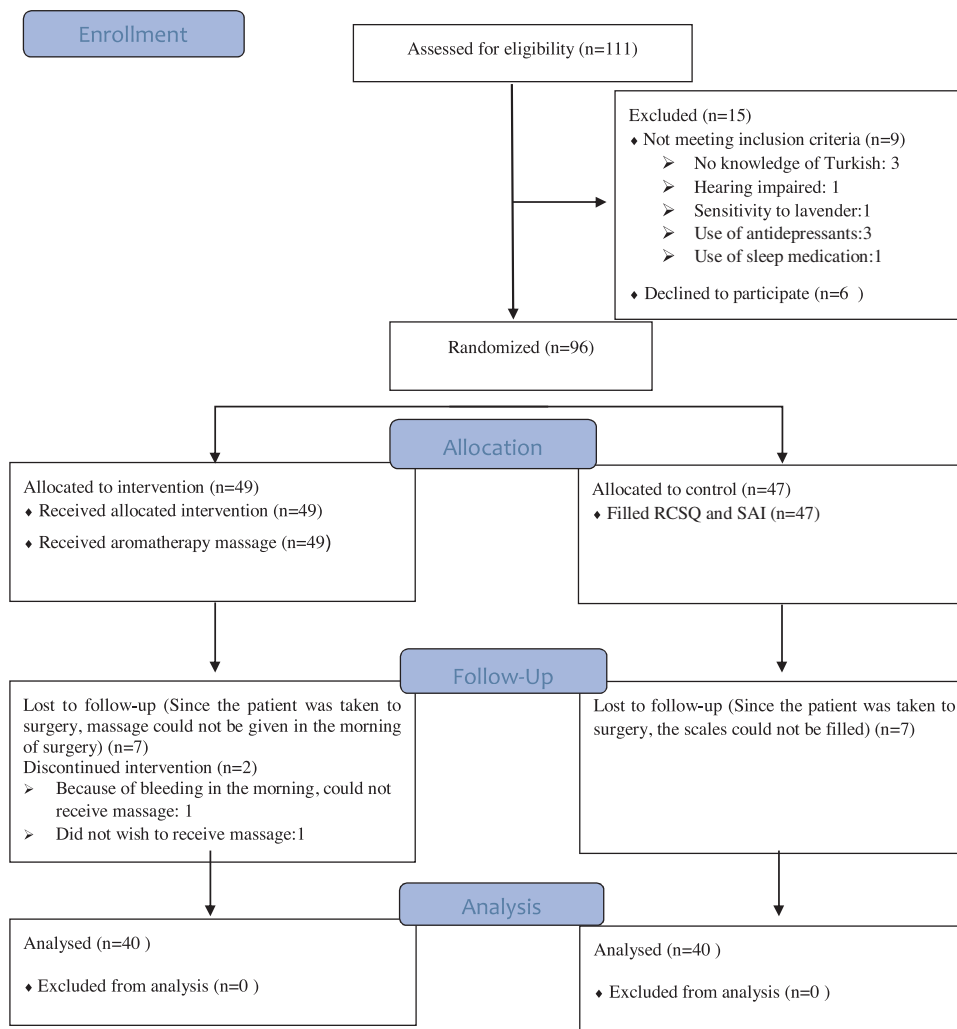
2.4. Outcome measures

In the research, Individual Identification Form, Richard-Campbell Sleep Questionnaire and State Anxiety Inventory were used for data collection. In accordance with the literature,^{13,24,28} the Individual Identification Form consisted of questions on the participants' age, gender, marital status, social security, educational status, working status, cigarette-alcohol consumption, other health problems, regular use of medication, previous surgeries undergone and the condition of having received training on the surgery. These questions were answered by the patients at the first meeting after they had agreed to participate in the research.

The preoperative anxiety level of colorectal surgery patients was obtained using the State Anxiety Inventory. The State Anxiety Inventory was developed by Spielberger et al. in 1970. A score of 0–19 points on the scale indicates that there is no anxiety while 20–39 points indicate mild, 40–59 points indicate moderate, 60–79 points indicate severe anxiety, and scores above 60 points indicate that individuals need professional help. The validity and reliability of the Turkish version of the inventory was made by Oner in 1977. The Cronbach α value for this study is 0.92.

The sleep quality of the patients was obtained with the Richard-Campbell Sleep Questionnaire (RCSQ). Developed by Richards in 1987, RCSQ is a five-item scale that assesses the perceived sleep depth, sleep

Chart 1. Randomization flow diagram.



latency (time to fall asleep), number of awakenings, efficiency (percentage of time awake) and sleep quality. Each item is evaluated on a 100-mm visual-analogue scale. Scores between “0–25” indicate very poor sleep while “76–100” indicate very good sleep. The cronbach α value of the scale is 0.82.²⁹ The validity and reliability of our scale in our country was made by Karaman Özlü and Özer³⁰ in 2015 and the cronbach α value is 0.91. The cronbach α value for this study is 0.92.

2.5. Validity and reliability

In this study, the randomized clinical trial was conducted in rigorous accordance with the CONSORT guidelines. First, a massage guide was developed for the intervention. The research assistants who conducted the interventions and collected data in this study followed the massage guide and performed the interventions at the same hour. To enhance the validity and reliability of the data, we used validated instruments with good reliability. Data collection was performed in accordance with the existing standard procedures. All the data obtained were checked carefully, right after collection in order to avoid any inaccurate information. Data were analyzed using advanced statistical analysis methods with contradictory factors also taken into consideration.

2.6. Ethical considerations

Written consents were received from the author of the Turkish Version of the RCSQ and the patients with the Voluntary Consent Form

after they were informed about the aim of the research as well as the Dokuz Eylül University Hospital and the Ethics Committee for Non-Interventional Research (Date of decision: 14.01.2015, Decision Nr: 2016/02-39).

2.7. Data analysis

The data obtained from the research were analyzed using the SPSS 15.0 program. Whether the continuous numeric variables had normal distribution was checked with Kolmogorov Simirnov, kurtosis, skewness and histogram. Descriptive statistics were made by giving the mean and standard deviation for variables with normal distribution.

Percentage was used to evaluate the descriptive characteristics of the patients while the chi-square significance test was used to compare the characteristics of the individuals in the experimental and control groups. T test was performed in independent groups in order to compare the mean scores of experimental and control groups from the Richard-Campell Sleep Questionnaire and the State Anxiety Inventory. A paired *t*-test was used to compare the mean scores of the experimental group obtained from the Richard-Campell Sleep Questionnaire and the SAI performed in the evening before surgery and in the morning of the surgery.

3. Results

A statistically significant difference was not found ($p > 0.05$, Table 1) between the experimental and control groups in terms of age,

Table 1
Descriptive characteristics of participants in the experimental and control group (n = 80).

Defining characteristics	Experimental group		Control group		Total		x ²
	(n = 40)		(n = 40)		(N = 80)		p
Age							
X ± SS	61.00 ± 11.35		60.28 ± 14.62		60.64 ± 13.01		t = 0.248 p = 0.805
	n	%	n	%	n	%	
Sickroom							
Single	17	42.5	12	30	29	36.3	x ² = 0.865 ^a
Multiple	23	57.5	28	70	51	63.8	p = 0.352
Gender							
Female	16	40	20	50	36	45	x ² = 0.455 ^a
Male	24	60	20	50	44	55	p = 0.500
Marital status							
Single	8	20	8	20	16	20	x ² = 0.000 ^a
Married	32	80	32	80	64	80	p = 1.000
Educational status							
Illiterate	2	5	1	2.5	3	3.8	x ² = 2.478 p = 0.649
Literate	3	7.5	1	2.5	4	5	
Primary education	12	30	16	40	28	35	
High school	13	32.5	10	25	23	28.7	
University	10	25	12	30	22	27.5	
Employment status							
Yes	6	15	7	17.5	13	16.3	x ² = 0.000 ^a
No	34	85	33	82.5	67	83.8	p = 1.000
Smoked							
Yes	4	10	6	15	10	12.5	x ² = 0.114 ^a
No	36	90	34	85	70	87.5	p = 0.735
Use alcohol							
Yes	2	5	4	10	6	7.5	p = 0.675 ^b
No	38	95	36	90	74	92.5	
Comorbid							
Yes	21	52.5	27	67.5	48	60	x ² = 1.302 ^a
No	19	47.5	13	32.5	32	40	p = 0.254
Constantly used medication							
Yes	21	52.5	28	70	49	61.3	x ² = 1.896 ^a
No	19	47.5	12	30	31	38.8	p = 0.169
Type of medication ^c							
Cardiovascular	9	22.5	19	47.5	28	35	x ² = 4.451 ^a , p = 0.35
Gastrointestinal	3	7.5	8	20	11	13.8	x ² = 1.686 ^a , p = 0.194
Central nervous system	3	7.5	4	10	7	8.8	p = 0.694 ^b
Hormone regulatory	8	20	10	25	18	22.5	x ² = 0.072 ^a , p = 0.789
Others	4	10	6	15	10	12.5	x ² = 0.114 ^a , p = 0.735
Previously undergoing surgery							
Yes	31	77.5	27	67.5	58	72.5	x ² = 0.564 ^a
No	9	22.5	13	32.5	22	27.5	p = 0.453
Education about surgery							
Yes	15	37.5	20	50	35	43.8	x ² = 0.813 ^a
No	25	62.5	20	50	45	56.3	p = 0.367

^a Since “n” is smaller than 25, Yates correction is made.

^b Fisher Corrected chi-square is used because of the expected count less than 5.

^c More than one medication type is marked.

gender, marital status, educational status, working status cigarette-alcohol consumption, other health problems, regular use of medication, previous surgeries undergone and the condition of having received training on the surgery.

3.1. Preoperative anxiety

When the preoperative mean scores taken from the State Anxiety Inventory were examined, the baseline measurement revealed no difference between the experimental and control group in terms of anxiety level (p > 0.05).

The experimental group who received aromatherapy massage had a mean SAI score of 42.25 ± 9.32 in the evening before surgery while this value decreased to 35.25 ± 6.80 in the morning of surgery and

that the difference was determined to be statistically significant (p < 0.05). In the control group, the mean preoperative SAI score was 42.67 ± 9.57 while this value increased to 45.40 ± 9.55 in the morning of surgery; it was found that the anxiety levels were higher and that this negative difference was statistically significant (p < 0.05) (Table 2).

3.2. Sleep quality

When the preoperative baseline measurements of RCSQ were examined, no difference was found between the experimental and control groups in terms of sleep quality (p > 0.05) (Table 2). When the table was examined, it was found that the preoperative RCSQ mean score of the experimental group who received aromatherapy massage was

Table 2

Distribution of the State Anxiety Inventory (SAI) and Richard Campbell Sleep Questionnaire (RCSQ) score averages according to experimental and control groups.

	Experimental group (n = 40)	Control group (n = 40)	t
	$\bar{X} \pm SS$	$\bar{X} \pm SS$	p
SAI score			
Preoperative evening	42.25 $\pm$ 9.32	42.67 $\pm$ 9.57	t = -0.201 p = 0.841
Morning of the surgery	35.25 $\pm$ 6.80	45.40 $\pm$ 9.55	t = -5.473 p = 0.000
t	t = 6.766	t = -2.704	
p	p = 0.000	p = 0.010	
RCSQ score			
Preoperative evening	46.02 $\pm$ 18.42	52.57 $\pm$ 23.17	t = -1.399 p = 0.166
Morning of the surgery	66.82 $\pm$ 17.98	42.80 $\pm$ 19.45	t = 5.735 p = 0.000
t	t = -7.219	t = 2.713	
p	p = 0.000	p = 0.010	

(46.02 $\pm$ 18.42), their sleep quality increased (66.05 $\pm$ 17.98) in the morning of the surgery and that this difference was statistically significant ($p < 0.05$). As for the control group, it was determined that sleep quality decreased compared to the preoperative evening (42.80 $\pm$ 19.45) and that this difference was statistically significant ($p < 0.05$) (Table 2). It was determined that the RCSQ mean score of the experimental group who received aromatherapy massage in the morning of surgery (66.82 $\pm$ 17.98) was significantly higher than that of the control group (42.80 $\pm$ 19.45) ($p < 0.05$).

Table 3 shows the distribution of the RCSQ domain mean scores to according to the experimental and control groups in the morning of surgery.

The mean scores of the experimental group taken from the RCSQ domains after aromatherapy massage in the morning of surgery were determined to be; sleep depth 64.37 $\pm$ 22.04, sleep latency 63.12 $\pm$ 21.26, number of awakenings 69.00 $\pm$ 21.39, efficiency 67.50 $\pm$ 22.41, and sleep quality 70.12 $\pm$ 19.46. The mean scores of the control group taken from the RCSQ domains were determined to be; sleep depth 42.62 $\pm$ 23.80, sleep latency 40.87 $\pm$ 22.52, number of awakenings 43.75 $\pm$ 22.86, efficiency 42.00 $\pm$ 19.92, and sleep quality 44.75 $\pm$ 22.27. As a result of the analysis, it was determined that there was a statistically significant difference between the Richard Campbell Sleep Questionnaire domain mean scores of the two groups in the morning of surgery (Table 3) ($p < 0.05$).

Table 3

Distribution of Richard Campbell Sleep questionnaire subscale scores according to experimental and control groups in the morning of surgery (n = 80).

RCUÖ Item	Experimental group (n = 40)	Control group (n = 40)	t
	$\bar{X} \pm SS$	$\bar{X} \pm SS$	p
Deep/light sleep	64.37 $\pm$ 22.04	42.62 $\pm$ 23.80	t = 4.240 p = 0.000
Sleep latency	63.12 $\pm$ 21.26	40.87 $\pm$ 22.52	t = 4.542 p = 0.000
Number of awakenings	69.00 $\pm$ 21.39	43.75 $\pm$ 22.86	t = 5.099 p = 0.000
Ease of return to sleep	67.50 $\pm$ 22.41	42.00 $\pm$ 19.92	t = 5.377 p = 0.000
Quality of sleep	70.12 $\pm$ 19.46	44.75 $\pm$ 22.27	t = 5.426 p = 0.000

4. Discussion

Aromatherapy, which is one of the complementary and integrative approaches widely used in many countries, is a part of the holistic nursing practice.¹⁷ In the literature review, it was found that there were many studies in which aromatherapy was performed with different sample groups^{14,22,23,31,32} there was a lack of studies examining the effect of lavender oil aromatherapy on anxiety in patients in the pre-operative period of colorectal surgery.

The finding of this study that aromatherapy massage reduces the level of anxiety in patients who will undergo colorectal surgery in the preoperative period demonstrates a similarity with other studies with oncology patient populations.^{22,32,33} It is also consistent with studies conducted with intensive care patients,³⁴ psychiatric patients,²³ and mothers of hyperactive children.¹⁴ It is known that anxiety is usually related to specific neural circuits and neurotransmitter systems. Stress affects major brain chemicals such as serotonin (regulating appetite, mood and sleep), dopamine (part of the brain reward system), noreadrenaline (regulation of energy and energy use) γ -amino butyric acid (GABA: general sedative effect), glutamate corticotrophin releasing factor (CRF: increasing steroid rate). It is known that lavender reduces exogenous emotional stimuli by increasing inhibitor neurons containing gamma amino butyric acid (GABA).³⁵ Furthermore, it is also known that massage can increase vagal activity by pressure stimulation and decrease cortisol level, lead to the development of immunity function and reduce stress.³⁶ At the same time, the rubbing action performed during the massage helps the absorption of essential oils used in aromatherapy and reveal their therapeutic effect. It is also thought that by revealing the scents, it also allows the individual to inhale the pleasant odors and contributes to the recovery.^{17,27} The olfactory memory is located on the hippocampus in the limbic system and the chemicals of the learned relaxing odors are processed in this area. This is how individuals are ensured to feel better when they meet the already-learned pleasant odors again.^{17,35} In our study, the fact that lavender oil was applied with massage suggests that it is effective in decreasing the level of anxiety.

Given that sleep has a very important place in maintaining the psychological and physiological well-being of individuals, inadequate and poor sleep has negative effects on, stress condition, immunity system, wound healing and cognitive functions. Lavender oil has been reported to be an excellent essential oil used to increase sleep quality of patients and to avoid sleep deprivation.³⁷ However, there is a limited number of studies on the effect of aromatherapy massage on sleep quality in the literature.^{15,38–40} Findings from this study show that aromatherapy massage applied with lavender oil improves sleep quality which is compatible with the literature.^{15,38,39} In parallel with the results of this study, Soden et al.¹⁵ reported that a 30-min back massage applied with 1% lavender oil improved sleep quality in their study conducted with palliative cancer patients. Similarly, studies conducted with hypertensive patients³⁹ and elderly patients⁴¹ revealed that lavender oil had a sedative effect and increased sleep quality, which supports the results of this study.

When the findings of this study and other studies are evaluated, it can be considered that the results obtained stem from the chemical structure of lavender oil and the massage practice of aromatherapy. Lavender, whose main chemical components are lavandulol, linalool, linalyl acetate, camphor, cineole, caryophyllene, camphor, dipentene, limonene, osmotic, and terpinene, has high ester content with sedative and relaxing properties.⁴² Lavender modulates intracellular cyclic adenosine monophosphate (cAMP) activity⁴³ and linalool, its principal component, inhibits the binding of glutamate. Linalool reduces the release of acetylcholine and alters the ion channel function at the neuromuscular junction.³⁷ Although these two factors are considered to have sedative effects the neuroanatomical pathways mediate lavender's sedative sleep effects but their mechanism remains unknown. However, all odors are thought to activate the amygdala in the cortex. These

odors are thought to send information to various regions of the brain including the sleep-wake cycle control.³⁷ It is also thought that massage can provide an increase in somatostatin release and deep sleep due to the inhibition of substance P.²⁷

According to the findings obtained from this study, it was determined that the perceived sleep depth, sleep latency, number of awakenings, efficiency, sleep quality domain mean scores were higher in the experimental group who received aromatherapy massage in the morning of surgery compared to the control group while their sleep quality domain was at a moderate level (Table 3). In their study which tested the inhalation of lavender oil in patients in the intermediate care unit, Lytle, Mwatha, and Davis⁴⁴ reported that the mean sleep scores for depth of sleep, ease of falling asleep, and quality of sleep were higher in the experimental group than in the control group. Similarly, Williams⁴⁰ reported that aromatherapy massage with lavender oil in autistic children was not effective on children's sleeping time, number of awakenings and sleep duration. The differing results of the study of Williams may have been caused by the fact that the study was conducted with a group of autistic children.

Exclusion criteria were determined to prevent the impact of other independent variables on the effect of aromatherapy massage and the differences between the groups. Patients with cognitive impairment, delirium or delusion were not contacted and users of certain drugs as mentioned in the literature have been excluded because they are thought to affect sleep quality (e.g. antihistamines increase sleep amounts). The individuals who used other complementary methods were excluded from the study since the progress in their results would be caused by other methods. For this reason, this sampling method limits the possibility of generalizing this study for a greater sample. Furthermore, preoperative anxiety is quite common and patients have a limited supportive interaction with physicians and nurses in the period before surgery. For this reason, massage given by the researcher might have affected the level of anxiety and constitutes a limitation of the study.

Another limitation of the study is the separation of patients into experimental and control groups with alternate enrollment. In this study, the patients were divided into groups according to the order of the surgery list before meeting the patient in order to avoid taking sides. Nevertheless, the use of randomization methods such as block randomization is recommended for future research.

The inability to blind the application to the researcher may be conceived another limitation of the design. This study could not be planned as a blinded trial as body odor is released while rubbing when the lavender oil is applied and the person receiving the massage is informed about the intervention. The impact of this limitation was tried to be reduced not by the observation of the researchers but rather through the information given by the patients regarding their own levels of anxiety and sleep. On the other hand, there is a limitation in the experimental design of this study that is the individual effects of the massage treatment and aromatherapy may not be distinguished. A minimum of four separate groups are required to enable the identification of the individual effects of massage therapy and aromatherapy; (1) massage applied alone (massages with inert carrier oil), (2) aromatherapy and massage applied together (massages with essential oil), (3) inhalation aromatherapy applied alone (without massage), and (4) control (neither massage nor aromatherapy application). It should be noted that this study lacks an equivalent placebo-control group to predict an expectation effect against aromatherapy massage group.

5. Conclusion

In colorectal surgery patients, lavender oil aromatherapy massage reduces the level of anxiety and increases the sleep quality in the preoperative period. In order to reduce anxiety and increase sleep quality as well as patient satisfaction in the preoperative period, lavender oil massage may be recommended.

Relevance to clinical practice

Nurses have certain responsibilities to ensure that patients who undergo colorectal surgery receive a successful operation, provide patient satisfaction, and maximize patients' well-being. Aromatherapy massage reduces the level of anxiety and increases the sleep quality in the preoperative period. In order to increase patient satisfaction and improve the results, aromatherapy massage may be recommended in the preoperative period. In-service training programs may be organized to encourage nurses to use complementary-integrative approaches and aromatherapy massage within the holistic care of patients.

Contributions

The authors have confirmed that all of the authors meet the IC-MJE criteria for authorship credit (www.icmje.org/ethical_1author.html), as follows: (CA, DÖ) making substantial contributions to the conception or design of the work; (CA) patient recruitment, data analysis and manuscript writing; (DÖ) drafting the article or revising it critically for important intellectual content.

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Conflict of interests

None.

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